

Things You Wanted to Know!

Q1. On every mobile charger, there is an instruction that says the charger can work on 100V-240V. Does it mean that the same charger will work in India and the USA too?

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A1. Every country has different rules regarding distribution of electricity for portable appliances and lighting. Voltage, frequency and plug type vary widely. Physical compatibility of receptacles may not ensure

(230V, 50Hz/120V, 60Hz) is rectified and smoothened by a reservoir capacitor before being processed to produce a regulated DC output at the required level.

Power supplies that use active power factor correction (PFC) usually are auto-ranging, supporting input voltages from ~100V AC to 250V AC, with no input voltage selector switch.

In the case of SMPS, a boost converter is inserted between the bridge rectifier and main input capacitors. The boost converter attempts to maintain a constant DC bus voltage on its output, while drawing a current that is always in phase with, and at the same frequency as, line voltage.

Another switched-mode converter inside the power supply produces the desired output voltage from the DC bus.

This approach requires additional semiconductor switches and control electronics but permits cheaper and smaller passive components. Due to their very wide input voltage range, many power supplies with active PFC can automatically adjust to operate

on AC power from about 100V in Japan to 230V in Europe and India, and 120V in the USA. This feature is particularly welcome in power supplies for laptops.

The figure above shows an example of the block diagram of a typical SMPS with an AC mains (line) input

and a regulated DC output. Output rectification and filter are isolated from the high-frequency switching section by a high-frequency transformer, and voltage control feedback is via an optoisolator.

The control circuit block is typical of specialist ICs containing a high-frequency oscillator, pulse width modulation, voltage and current control, and output shut-down sections.

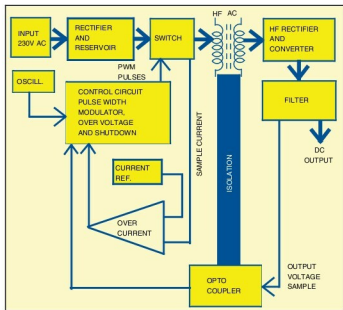
Q2. I have a circuit that can control either the upper tank or the lower tank. I need a circuit that is capable of controlling both upper and lower tanks. Please help.

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A2. The circuit of 'Numeric Water Level Indicator' was published in February 2010 issue of EFY. Most water-level indicators are based on the number of LEDs that glow to indicate its level.

This water-level indicator uses one 7-segment display to show nine different levels of water in the tank. It is built around priority encoder IC 74HC147, BCD-to-7-segment decoder IC CD4511, 7-segment display LTS543 and a few discrete components. If you are interested, you can get the kit from EFY associates, Kits'n'Spares (<http://kitsnspares.com/user1/buyproduct.asp>).

Another article 'Water Level Indicator Using 7-Segment Display' was published in May 2010 issue of EFY. The three water levels, low (L), high (H) and full (F), are displayed in three different 7-segment displays. A buzzer is provided to alert the user in case of water overflowing from the tank.



Block diagram of a typical SMPS

compatibility of voltage, frequency or connection to earth ground. India and the USA follow plug standard IS 1293 (230V, 50Hz) and NEMA 1-15 P (120V, 60Hz), respectively.

Power adaptors for laptops and mobile chargers are switched-mode power supplies (SMPSes). AC mains

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